

# Apriori Algorithm - Theory & Explanation

## Aim

To understand and apply the Apriori algorithm for finding frequent itemsets and generating association rules in transactional data.

## Theory

The Apriori algorithm is one of the classic algorithms used in association rule mining. It is mainly applied to market basket analysis, where the goal is to find patterns such as: 'If a customer buys bread and butter, they are likely to buy milk.'

## Key Concepts

1. Frequent Itemsets: An itemset is frequent if it appears in the dataset with support greater than or equal to a minimum threshold.
2. Association Rules: Rules have the form  $A \rightarrow B$ . Metrics:
  - Support: Proportion of transactions containing the itemset
  - Confidence: Likelihood of B appearing when A appears
  - Lift: Indicates strength of rule over random co-occurrence

## How the Apriori Algorithm Works

1. Start with single items meeting minimum support threshold.
2. Generate combinations (itemsets of size 2, then 3, etc.).
3. Prune itemsets that do not meet support threshold.
4. Continue until no further frequent itemsets can be found.
5. Generate association rules using frequent itemsets.

## Example

Transactions:

T1: Milk, Bread

T2: Milk, Diaper, Beer, Eggs

T3: Milk, Bread, Diaper, Beer

T4: Bread, Milk, Diaper, Beer

T5: Bread, Milk, Diaper, Coke

Example rule:

{Milk, Diaper} -> {Beer} with high confidence

## **Applications**

- Market Basket Analysis
- Recommender Systems
- Inventory Management
- Cross-selling strategies

## **Conclusion**

The Apriori algorithm is a foundational technique in association rule learning. It efficiently finds relationships among items in large datasets using support and confidence thresholds. While it can be computationally expensive, it remains a widely-used and intuitive method for discovering hidden patterns.